BitGuard

Bitcoin Address Risk Intelligence

MIDS Final Capstone Presentation
Section 6

Mission

Build a data-driven system that identifies risky Bitcoin wallets before fraud occurs, helping exchanges, businesses, and individuals transact confidently and safely in the crypto ecosystem.

BitGuard

Team Introduction

Wes Morberg

Modelling/Backend

Steven Au

Infra/Full Stack Lead

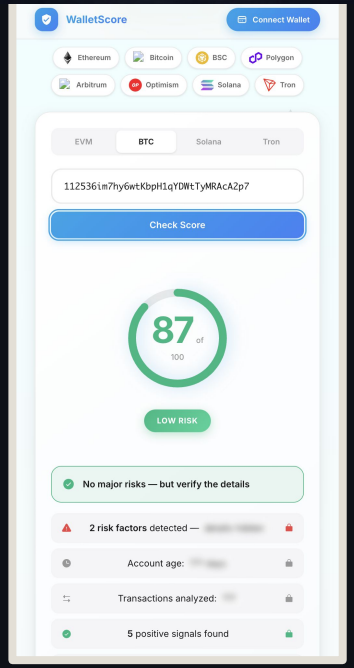
Noah Cederholm

Front End/EDA

Chirag Agarwal

Data Pipeline/Project

Crypto Scams are a Massive Barrier to Adoption



WHY THIS PROBLEM MATTERS

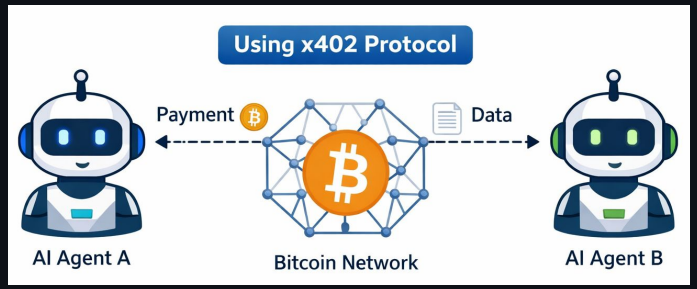
- Unlike traditional finance, Bitcoin is anonymous and irreversible
  - Over $17B in crypto losses in 2025 alone due to fraud[1]
  - ~50% of transactions come from new users - unaware of scam risks[2]
  - AI agents are emerging as a key driver in prediction markets - $6 billion in weekly volume in March 2026[3]
- Existing crypto risk tools are either too unreliable or locked behind enterprise-level paywalls
  - WalletScore →

1. https://www.chainalysis.com/blog/crypto-scams-2026/
2. https://s27.q4cdn.com/397450999/files/doc_financials/2024/q4/Q4-24-Shareholder-Letter.pd
3. https://www.coinbase.com/bytes/archive/the-future-of-ai-and-crypto?utm_source=chatgpt.comf

Target Users & Pain Points

Retail Users:

AI Agents:



PAIN POINTS WE HEARD

- Users want fast clarity, not deep investigation

- Trust comes from a score with a simple explanation

WHAT WE BUILT FOR

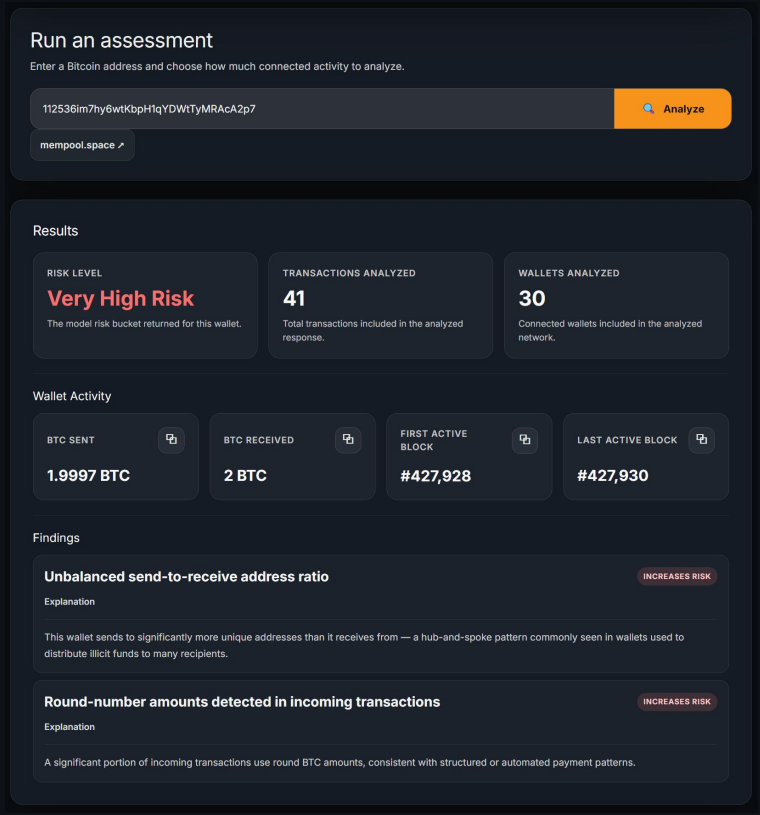
- Explainability had to be part of the MVP.

-The interface had to feel lightweight and immediately understandable.

- Output needed to support a real send / do not send decision.

MVP Demo: What the User Experiences

Website: https://bitguard.pro/



DEMO STORY

1. User pastes a Bitcoin address into BitGuard
2. The API returns:
   - a risk category from a confidence signal
   - plain-language explanations for why the risk was identified
   - Attributes of the address and transactions, including a link to mempool.space
3. The user can use that output to decide whether to continue, or investigate further.

HOW THIS SOLVES PAIN POINTS

- Fast response (typically under 5 seconds)
- Simple explanations of risk
- Stateless for privacy considerations

Technical Approach Overview

1. Data Ingestion

Known-address labels, transaction graph data, address-level history

2. Feature Layer

Behavioral, graph, transactional, and reputation-derived signals based on transaction connections

3. Model Layer

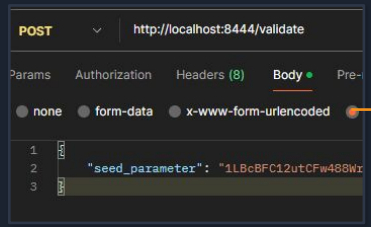
Train candidate classifiers to estimate address risk

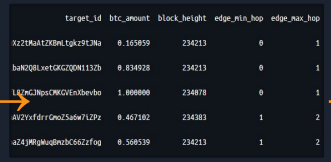
4. Explanation Layer

Map model outputs back to interpretable reasons / flags

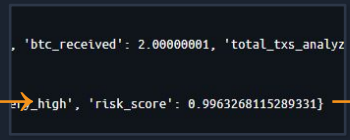
5. Product Layer

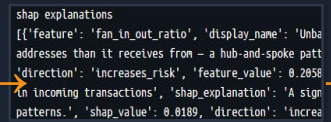
Surface a simple risk result through a user-facing app or API

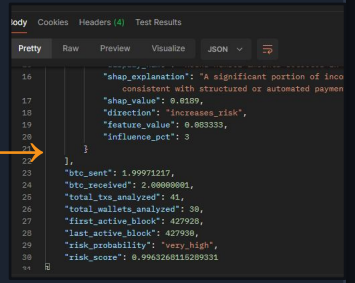




target_id	btc_amount	block_height	edge_min_hop	edge_max_hop
Xz2tMaAtZKBmLtgkz9tJNa	0.165859	234213	0	1
baNZQ8LxetcXGZQDNH113Zb	0.834928	234213	0	1
LB7mGJNpsCMKGVEnXbevbo	1.000000	234878	0	1
AVZYxfdrrGmoZSaGw7LZPz	0.467102	234383	1	2
qZ4jMRgWuqBnzbCG6Zzfog	0.560539	234213	1	2







Exploratory Graph Data Analysis and Pipeline

KEY EDA TAKEAWAYS

- Graph needed over Relational Database
  - allows for quick exploration
  - Tried 3 hops and dynamic hops
  - Settled on 2-hops

- Supernode Avoidance
  - -5000 ms + 1000 Tx limit
  - avoids searching Crypto Exchanges
  - focuses on wallet → wallet

- Major Hurdles in compute to parse all data
  - 2 Tb Graph
  - ultimately decided to move forward with full graph to save time

DATASET DESCRIPTION [1,2]

~2.4 Billion Nodes
~26.3 Billion Edges
-16-year history from inception

- Multiple Feature Families
  - Transactional
  - Structural
  - Temporal
  - reputation-based

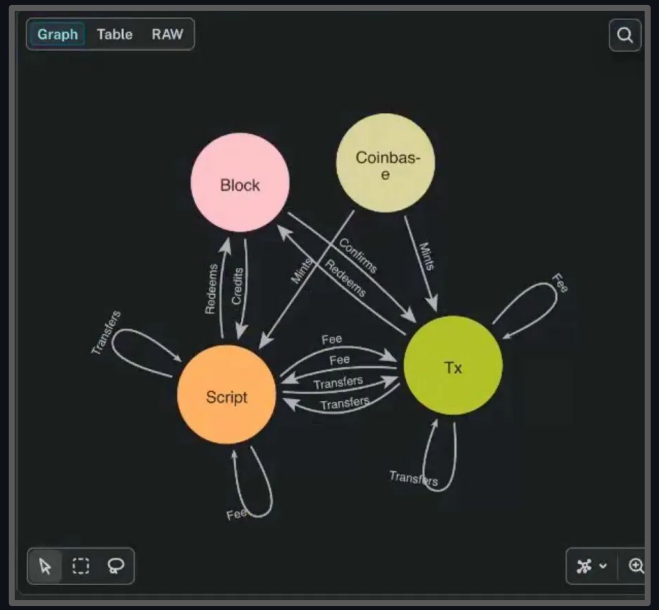
Database information

Nodes (2,394,735,857)

* Block Coinbase Script Tx

Relationships (26,346,298,498)

* Confirms Credits Fee Mints Redeems Transfers



1. https://eba.b1aab.ai/
2. https://arxiv.org/pdf/2510.20028

Training Data Sources: BitcoinHeist and Elliptic++

BITCOINHEIST [1,2]

- Academic dataset of Bitcoin ransomware addresses developed at UCI covering 28 ransomware attack families

2.9m	~29k	28
raw address records	illicit after dedup	ransomware families

ELLIPTIC++ [3]

- Graph-based dataset of illicit Bitcoin addresses developed at Georgia Tech, covering a broad range of illicit wallet categories including darknet markets, exchange hacks, and scams

822k	~6k	6
wallet addresses	illicit after dedup	illicit categories

COMBINED

513k	35k
total wallets	illicit wallets
6.8%	13:1
illicit base rate	class imbalance ratio

WHY BOTH?

- Allows the model to train on a broader range of illicit wallets and their on-chain data

- The only two publicly available labeled Bitcoin wallet datasets with meaningful scale

1. Bitcoin Heist Ransomware Address [Dataset]. (2020). UCI Machine Learning Repository. https://doi.org/10.24432/C5BG8V.
2. https://arxiv.org/pdf/1906.07852
3. https://github.com/git-disl/EllipticPlusPlus

Feature Engineering and Key Signals

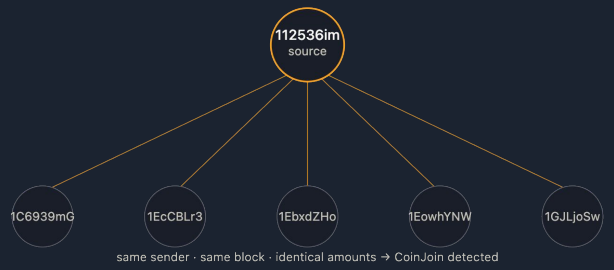
- 138 total features transformed from neo4j graph output, grouped by hop distance and direction

KEY SIGNALS

- Coinjoin activity: 7x more common in illicit wallets

SOURCE	TARGET	BTC	BLOCK
112536im	1C6939mG	0.1738334	427928
112536im	1EcCBLr3	0.1738334	427928
112536im	1EbxdZHo	0.1738334	427928
112536im	1EowhYNW	0.1738334	427928
112536im	1GJLjoSw	0.1738334	427928

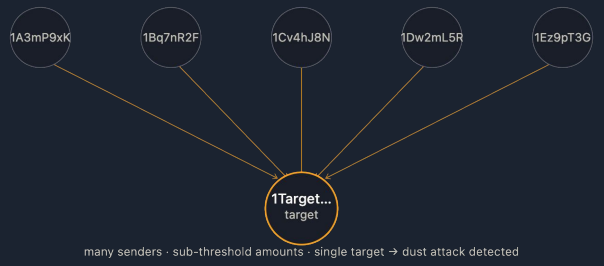
● same sender · same block · identical amounts → CoinJoin



- Dust transactions: 8x more common in illicit wallets

SOURCE	TARGET	BTC	BLOCK
1A3mP9xK	1Target…	0.00000546	774201
1Bq7nR2F	1Target…	0.00000612	774201
1Cv4hJ8N	1Target…	0.00000489	774202
1Dw2mL5R	1Target…	0.00000558	774202
1Ez9pT3G	1Target…	0.00000601	774203

● many senders · sub-threshold amounts · single target → dust attack



Modeling Approach and Selection

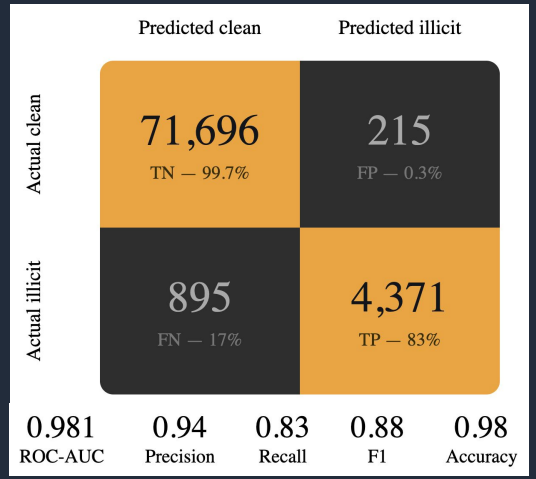
Goals: High Accuracy (F1) | High Interpretability | Fast Inference Time

MODEL CANDIDATES

Logistic Regression	Baseline model with high interpretability and very fast inference (< 0.1ms), limited ability to learn complex feature interactions
Random Forest	Baseline tree model (non-linear decision boundary, interpretable via SHAP score, fast inference (~0.3ms)
LightGBM	Optimized boosted tree model, interpretable via SHAP score, fast inference (~0.1ms), fast training
Graph Neural Networks (GNNs)	Best for graph data, limited interpretability ("black box"), slow inference (~100ms-1s)

FINAL MODEL

LightGBM



Why it won: best overall performance in the test set (n = 76,962), fast inference, results interpretable via SHAP scores

Model Performance and Limitations

Where the Model Performs Well

- High CoinJoin Tx activity in network

- High Dust Tx activity in network

- Suspicious hub-and-spoke patterns (sends to many unique recipients from 1 source)

- Short bursts of high activity, common in burner wallets

Edge Cases and Limitations

- Model skews toward ransomware patterns (dominant type in training data)

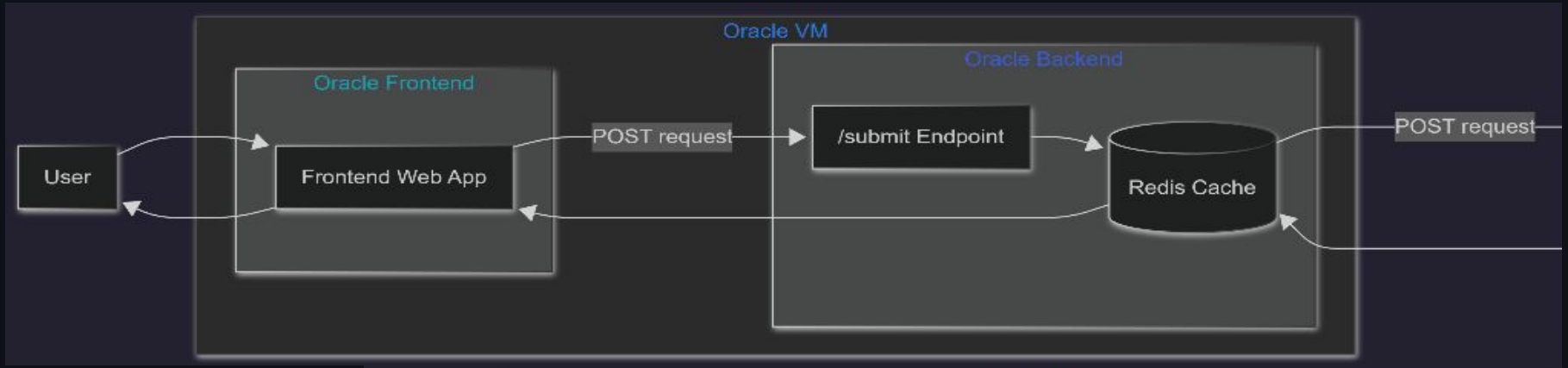
- Privacy-oriented wallets using Coinjoin legitimately (Wasabi, JoinMarket)

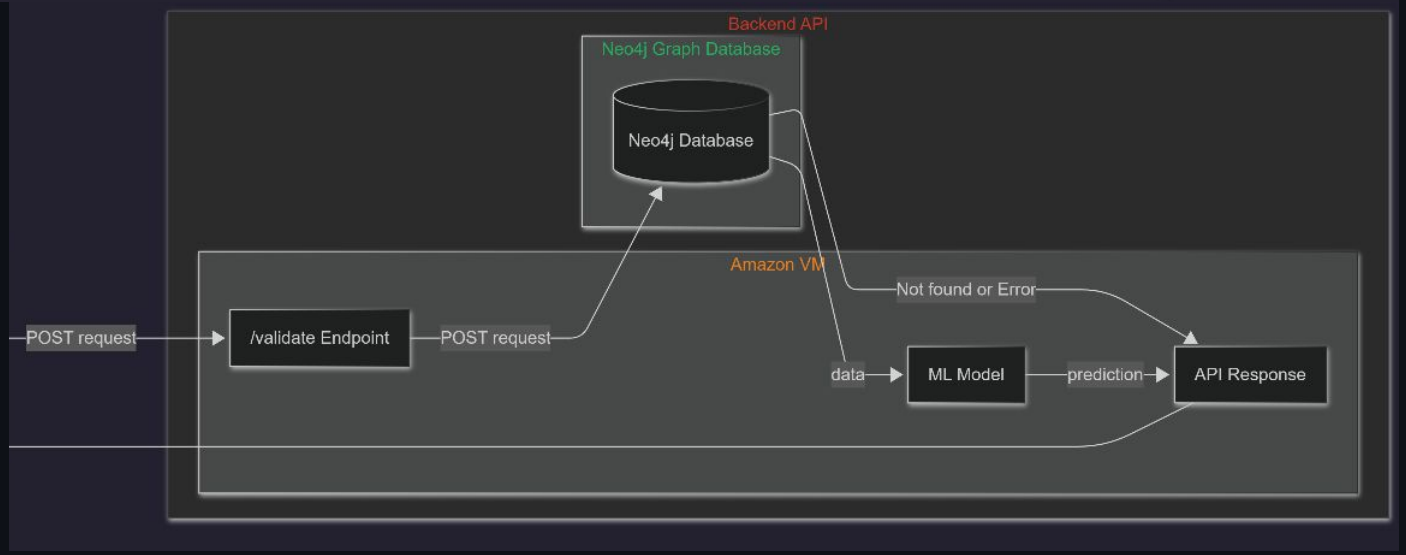
- Novel illicit on-chain patterns may be harder to detect if not learned by the model

RISK LEVEL

Mixed Signal

System Architecture Walkthrough





Key Technical Challenges

CHALLENGES

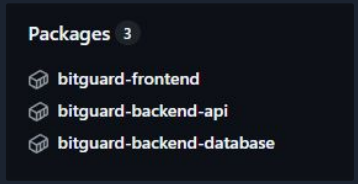
1. Monolithic Fragility
   - Frontend, cache, model API, and graph database as one entity issue.

2. Database Computation
   - Queries exponentially blew up on Supernodes

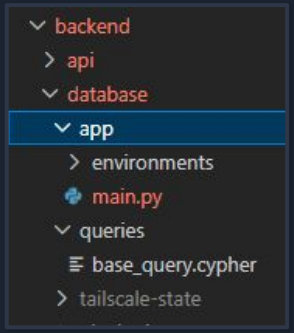
3. User Centric Design
   - Model API response output was too complex

SOLUTIONS

1. Separation of Concerns
   - Mutual Synergy (via JSON)
   - Monetizable API



2. Database Optimization
   - Optimized index for Cypher queries



3. Usability Engineered
   - Cached Results
   - Interpretable Frontend
   - Modern Mobile Support

Roadmap & Expansion

TOP NEXT STEPS

- Proper session token w/ x402 protocol[1] compatibility
    - Allows AI agent pay-per-call monetization with our API
- Ingest up-to-date transaction data
    - Allows the latest blocks to be trained on
- Train to catch additional malicious behaviors
    - Atomic Swaps
    - Additional Cryptocurrencies (Ethereum, Solana)
- UX improvements
    - User Profiles / Saved Searches / x402 private tokens
    - User Wallet Integration
    - Chrome Extension / Discord & Telegram Bots

x402

→ Accept payments with a single line of code

```
app.use(
  paymentMiddleware(
    {
      "GET /weather": {
        accepts: [...],                    // As many networks / schemes as you want to support
        description: "Weather data",       // What your endpoint does
      },
    },
  )
);
```

1. https://www.x402.org/

Addressable Market

- TokenSniffer[1] is a comparable product charging $100/month for Ethereum Tokens
  - Not graph/ML based
  - Token limited

- BitGuard has $30.7M ARR If we charge $20/month to 1% of US Cryptocurrency users

TOKENSNIFFER API

Sniffer Pack Pro

500 smell tests per day

$99.00 /mo

or $1128 /yr (5% discount)

Includes:

- ✓ Token score
- ✓ Scam status
- ✓ Test results with raw data
- ✓ 500 tokens per day
- ✗ Daily malicious tokens list
- ✗ Daily malicious addresses list

1. https://tokensniffer.com/

HOW THIS SCALES

55M holders × 31% transfer use case × 75% scam/security concern = 12.8M users

TOTAL ADDRESSABLE MARKET:

$1.4B–$2.5B ARR

40–70M active users × $240/year
Assumption: $ 20 / month consumer plan

SERVICEABLE AVAILABLE MARKET

$460M ARR

12.8M U.S. users × $240/year

SERVICEABLE OBTAINABLE MARKET

$30.7M–$153.6M ARR

1–5% of SAM = 128k–639k users

AI Agent wallet/exchange distribution can boost meaningful scale quickly.

Wrap-Up

Build a data-driven system that identifies risky Bitcoin wallets before fraud occurs, helping exchanges, businesses, and individuals transact confidently and safely in the crypto ecosystem.

Thank You!

ACKNOWLEDGEMENTS

Course Staff:
- Puya Vahabi
- Danielle Cummings

Team:
- Chirag Agarwal
- Noah Cederholm
- Steven Au
- Wes Morberg

BitGuard

ADDITIONAL RESOURCES

Github:
https://github.com/thesteau/bitguard

Website:
bitguard.pro